

Customer Shopping Behavior Analysis

Analyzing 3,900 customer transactions to uncover spending patterns, preferences, and strategic opportunities across product categories and customer segments.



Dataset Overview

3.9K

Total Transactions

Purchase records analyzed

18

Data Features

Columns tracking behavior

37

Missing Values

In Review Rating column

Key features include customer demographics (age, gender, location, subscription status), purchase details (item, category, amount, season), and shopping behavior (discounts, reviews, shipping preferences).



Data Preparation in Python

O1

Data Loading & Exploration

Imported dataset using pandas, checked structure with `df.info()` and summary statistics

O2

Missing Data Handling

Imputed Review Rating nulls using median rating per product category

O3

Feature Engineering

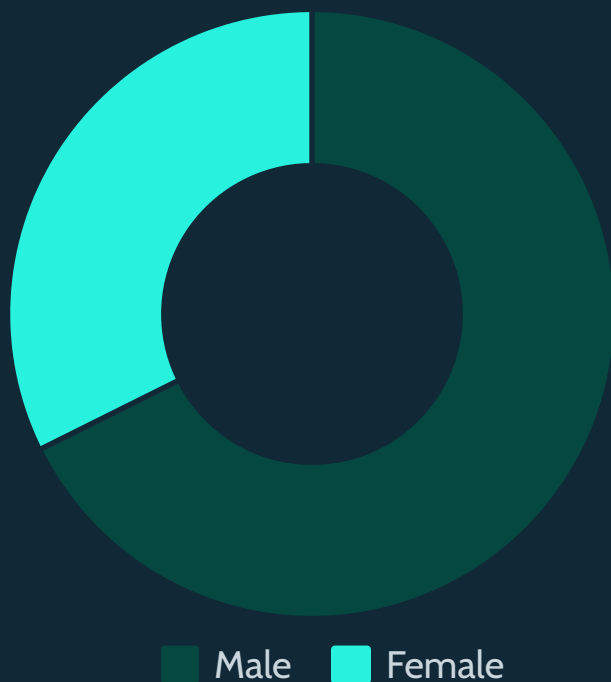
Created `age_group` bins and `purchase_frequency_days` columns for deeper analysis

O4

Database Integration

Connected to PostgreSQL and loaded cleaned data for SQL analysis

Revenue Insights by Gender



Male Customers Drive Revenue

Male customers generated \$157,890 in total revenue compared to \$75,191 from female customers, representing 68% of total sales.

This significant gap suggests opportunities for targeted marketing to female demographics.



Product Performance Analysis

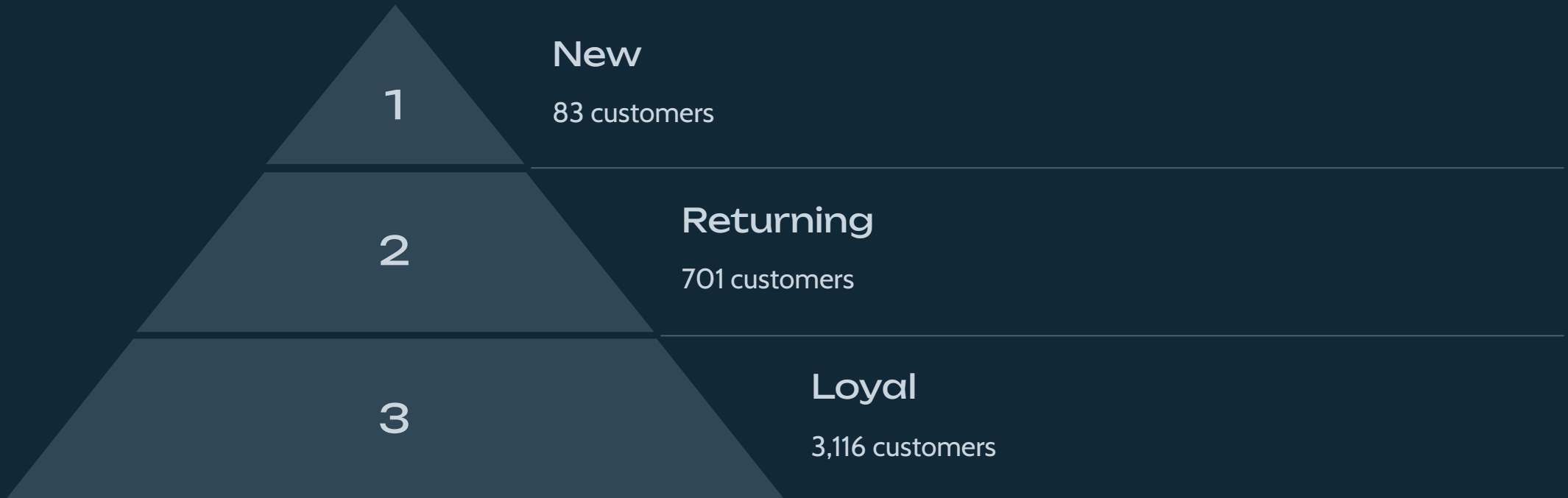
Top-Rated Products

1. Gloves (3.86 rating)
2. Sandals (3.84 rating)
3. Boots (3.82 rating)
4. Hat (3.80 rating)
5. Handbag (3.78 rating)

Discount-Dependent Items

1. Hat (50% discount rate)
2. Sneakers (49.66%)
3. Coat (49.07%)
4. Sweater (48.17%)
5. Pants (47.37%)

Customer Segmentation



Customer classification based on purchase history reveals a strong loyal base (80%), with opportunities to convert returning customers into loyal advocates through targeted engagement strategies.

Subscription & Loyalty Patterns

Subscription Comparison

- Non-subscribers: 2,847 customers, \$59.87 avg spend, \$170,436 total revenue
- Subscribers: 1,053 customers, \$59.49 avg spend, \$62,645 total revenue

Similar average spend but lower subscriber count indicates untapped potential.

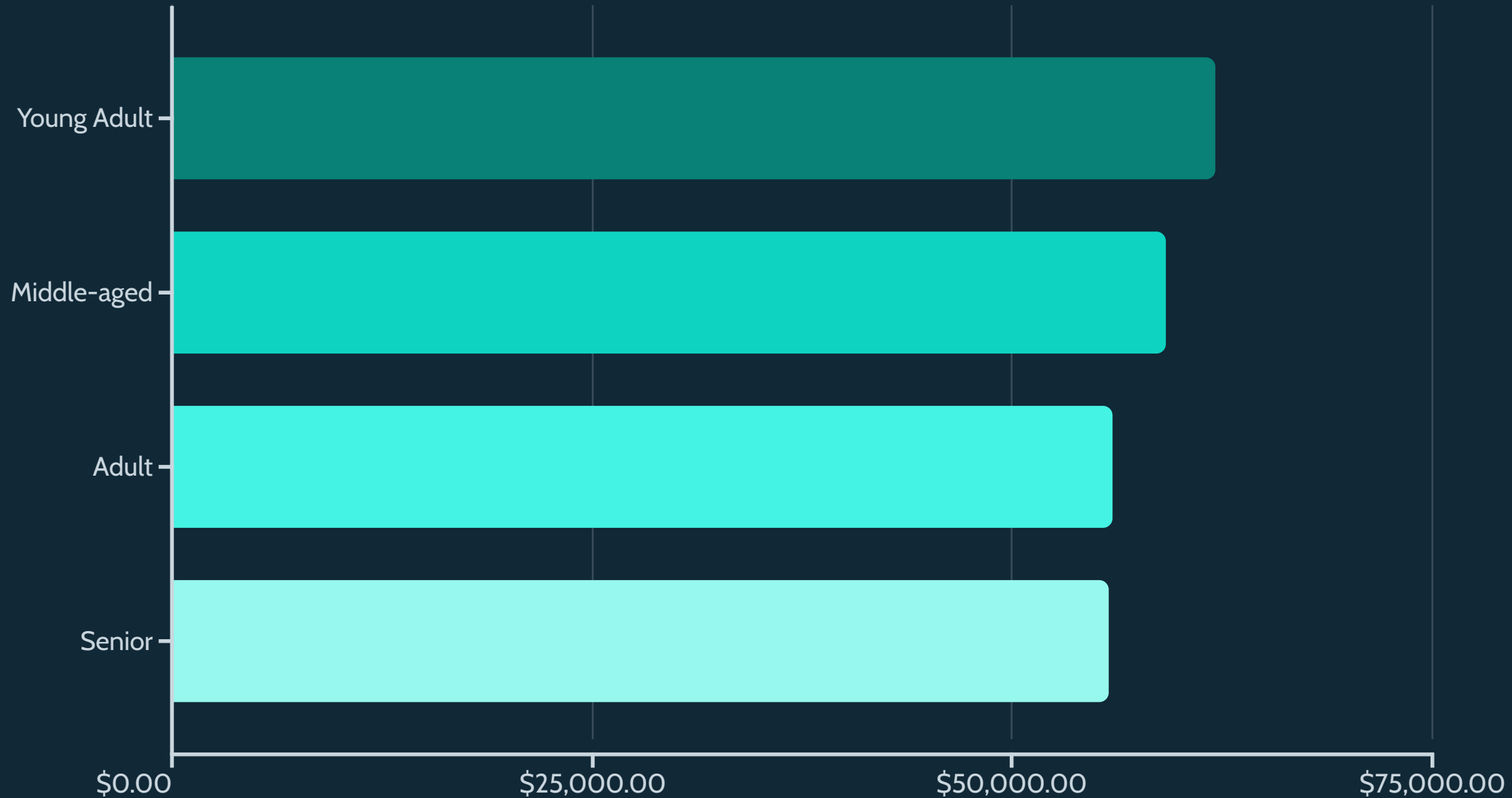
Repeat Buyer Insight

Among customers with 5+ purchases:

- 958 are subscribers
- 2,518 are non-subscribers

High-frequency buyers show strong subscription conversion opportunity.

Revenue by Age Group



Young adults lead revenue generation at \$62,143, followed closely by middle-aged customers. Revenue distribution is relatively balanced across age groups, suggesting broad market appeal.

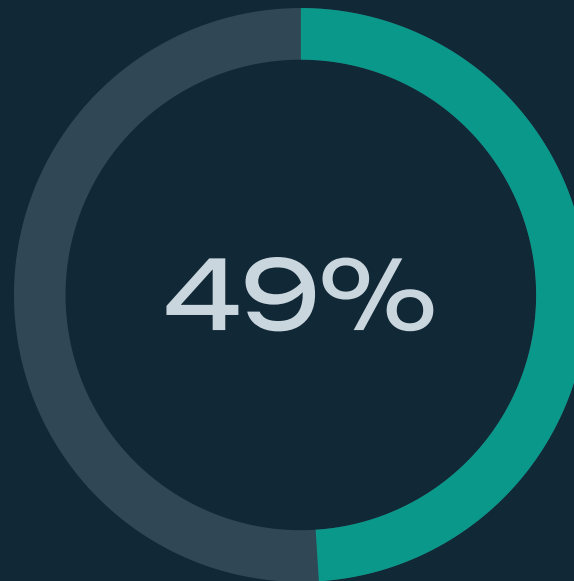
Shipping & Discount Behavior

Express vs. Standard Shipping

Express shipping customers spend \$60.48 on average, compared to \$58.46 for standard shipping—a modest 3.5% premium.

High-Spending Discount Users

Multiple customers used discounts while spending above average, indicating discounts don't necessarily reduce basket size.



Average Discount Rate

Across top discount-dependent products

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits to convert high-frequency buyers into subscribers



Customer Loyalty Programs

Reward repeat buyers to accelerate movement into loyal segment



Review Discount Policy

Balance promotional sales boosts with margin control strategies



Targeted Marketing

Focus on high-revenue age groups and express-shipping users for maximum ROI

